

Welcome

Chocolate Sales Prediction Using Linear Regression

Course : Artificial Intelligence and
Expert System

Presented to
Fawzia Fairouz

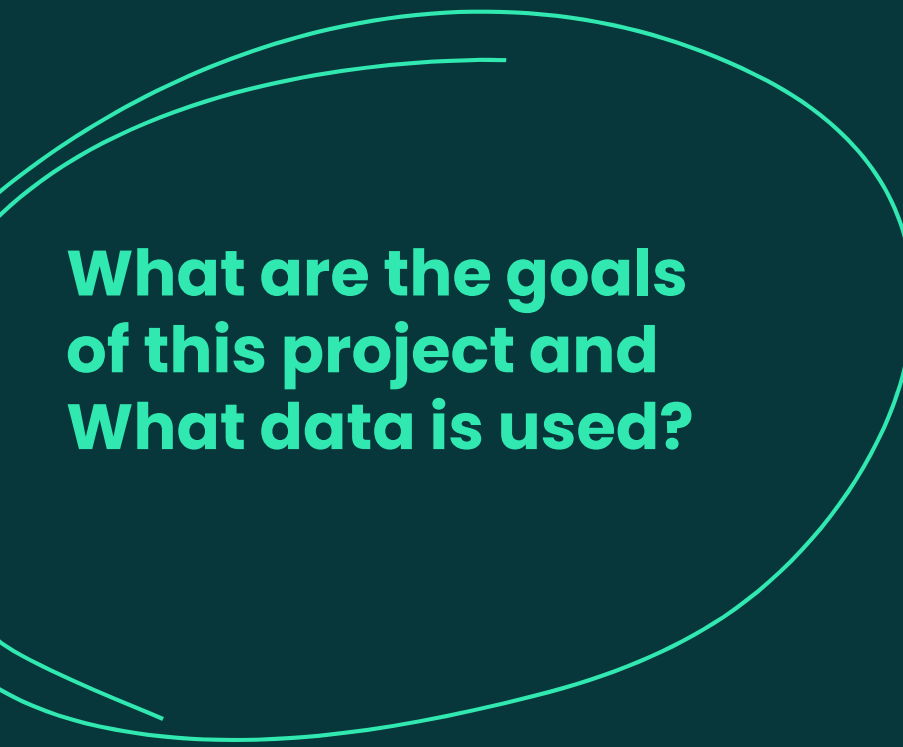
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Project Overview

Chocolate Sales Prediction Using Linear Regression

This project analyzes chocolate sales data to identify sales patterns and predict future sales amounts using a Linear Regression model. The analysis is based on product, country, and date information, helping to understand sales trends and improve prediction accuracy.



**What are the goals
of this project and
What data is used?**

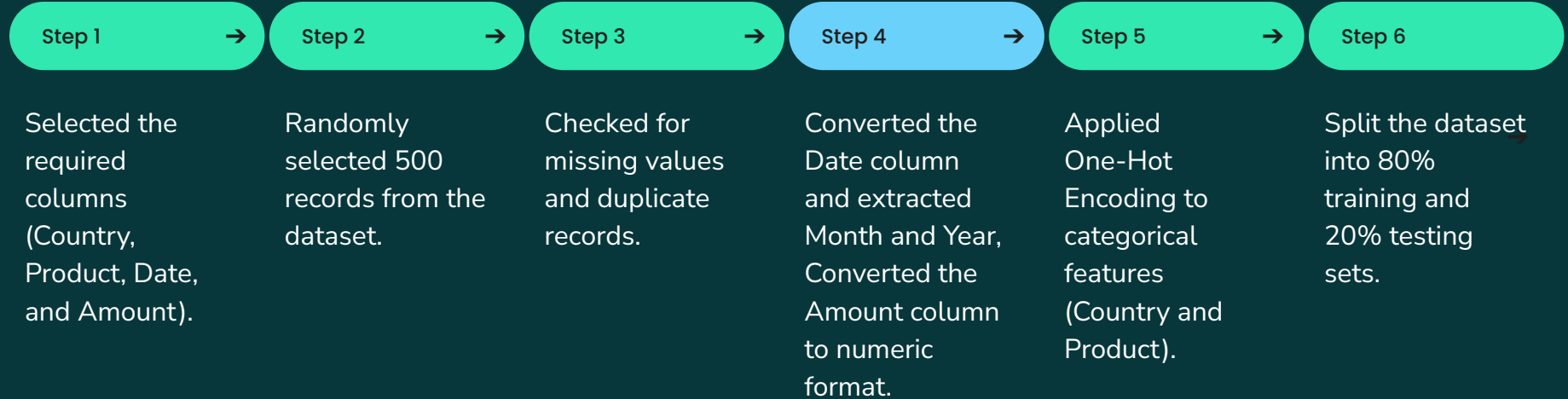
Objectives

- Analyze chocolate sales data.
- Clean and preprocess the dataset.
- Build a Linear Regression model.
- Evaluate the model's performance.
- Predict future sales amounts.

Dataset Features

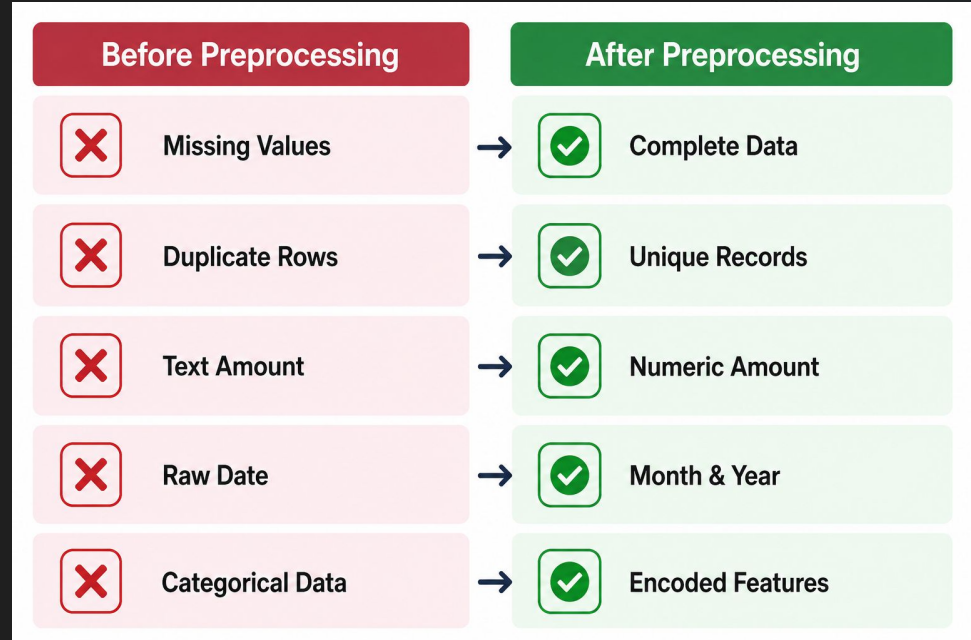
- Country
- Product
- Date
- Amount (Target Variable)

Data Preprocessing – Steps



Data Preprocessing – Purpose and Importance

Data preprocessing improves the quality of the dataset before training the model. It removes inconsistencies, converts data into a suitable format, extracts useful features, and prepares the dataset for accurate training and testing, resulting in more reliable predictions.



Exploratory Data Analysis (EDA)

What insights were obtained from the data?

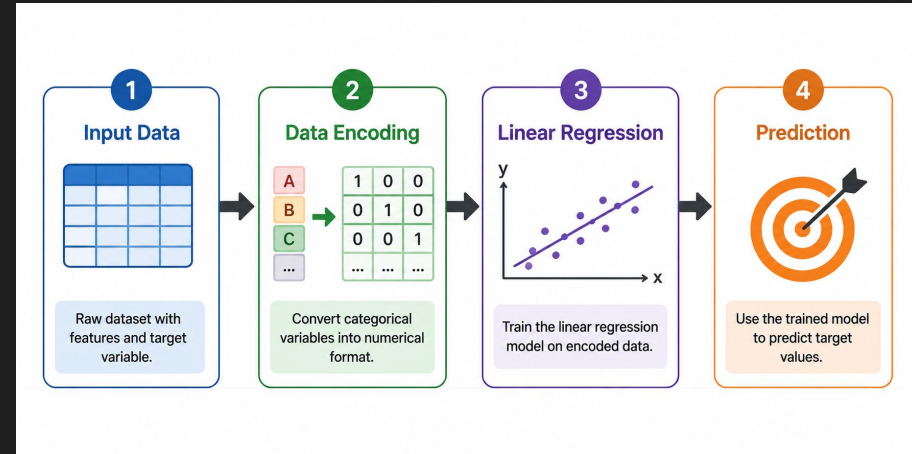
- Analyzed the distribution of sales amounts.
- Detected outliers using a boxplot.
- Compared total sales by country.
- Compared total sales by product.
- Identified the highest and lowest sales records.
- Calculated the average sales by country and product.
- Visualized the results using histograms, box plots, and bar charts.



How was the prediction model developed?

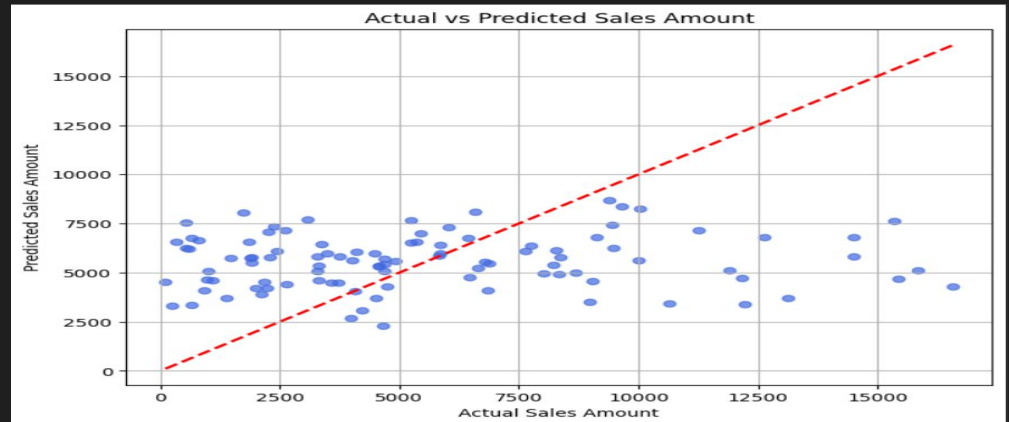
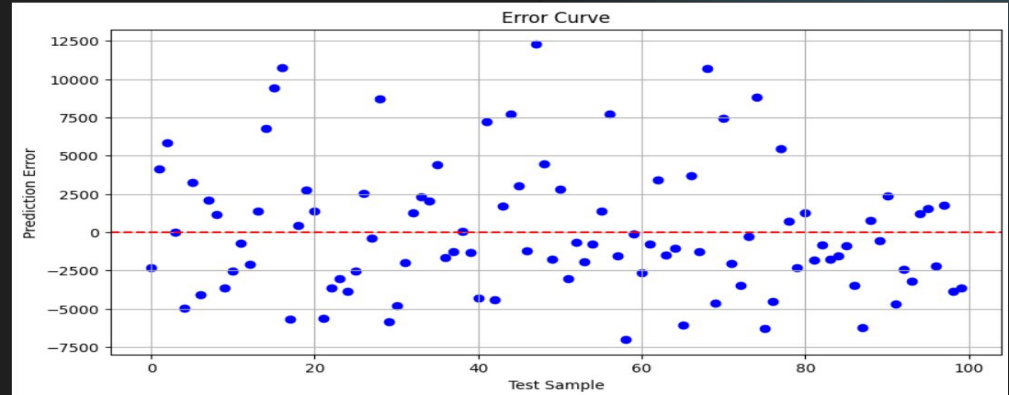
How was the prediction model developed?

- Selected **Country, Product, Month, and Year** as input features.
- Used **Amount** as the target variable.
- Split the dataset into **80% training** and **20% testing** sets.
- Applied **One-Hot Encoding** to categorical features.
- Trained a **Linear Regression** model.

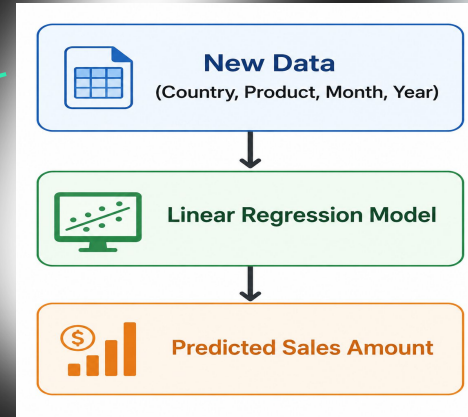


How was the model evaluated?

- Predicted sales amounts using the trained model.
- Evaluated performance using:
 - Mean Absolute Error (MAE)
 - Mean Squared Error (MSE)
 - Root Mean Squared Error (RMSE)
 - R^2 Score
- Compared actual and predicted values.
- Visualized prediction errors using an Error Curve.



How was the model used for prediction?



- Used the trained **Linear Regression** model to predict sales amounts.
- Provided **Country, Product, Month, and Year** as input features.
- Generated the predicted sales amount for new data.
- Demonstrated the model's ability to estimate future sales based on historical data.

Country	Product	Month	Year	Actual Amount	Predicted Amount	Error
India	White Choc	3	2023	2183.68	4525.055052	2341.375052
UK	Almond Choco	6	2023	11258.39	7147.801593	-4110.588407
UK	Raspberry Choco	1	2022	12635.00	6786.803744	-5848.196256
UK	Smooth Silky Salty	7	2023	5828.22	5842.821866	14.601866
UK	Peanut Butter Cubes	4	2023	2362.01	7333.697887	4971.687887
Canada	Organic Choco Syrup	1	2024	9479.63	6243.383875	-3236.246125
Australia	Organic Choco Syrup	1	2022	994.00	5077.690690	4083.690690

THANK YOU!

Conclusion :

- Successfully analyzed and preprocessed the chocolate sales dataset.
- Built a **Linear Regression** model for sales prediction.
- Evaluated the model using **MAE, MSE, RMSE, and R² Score**.
- Demonstrated the ability to predict future sales from historical data.